

Implementation Status

What was built

The repository has been normalized into the documented layout:

README.md	Setup and run instructions
brief.yaml	Campaign input
prompts/00...05.md	Runtime-loaded stage instructions
docs/	ICP, strategy, walkthrough, failures, this status
agent/run.py	CLI orchestration and artifact writing
agent/llm.py	OpenAI, OpenRouter, and Anthropic provider adapter
agent/tools.py	Run event, trace, failure logging, and fetch cache
agent/schemas.py	Pydantic contracts for stage outputs
agent/stages/	One module for every pipeline stage
agent/qa.py	Deterministic email guardrails
tests/	Offline schema/provider/guardrail tests

brief.yaml, the prompts, and the supplied strategy documents were preserved as the source of truth. Each stage module loads its corresponding file from prompts/ at runtime; no prompt text is embedded in Python.

Pipeline implementation

Stage	Implementation	Artifact on successful run
0: ICP	Live provider web search plus Pydantic validation. The fixed scoring rubric is parsed from docs/ICP.md at runtime.	icp_profile.json
1: Accounts	Live web-search model call, source-bearing fit reasons, score/tier filtering, JSON and CSV writers.	accounts.json, accounts.csv
2: Contacts	Live public-source discovery for Ops, HSE, and Site roles; NOT_FOUND remains valid output.	contacts.csv
3: Research	Per-account Markdown research brief through live search.	research/<account>.md
4: Email	Per-contact draft grounded in the research brief.	emails/<contact>.md, emails.json
5: QA	A separate model call plus deterministic checks; up to two rewrites.	qa_report.json

Account research tasks are concurrency-limited with --concurrency. Every live-search prompt and every source URL returned in structured output is written to trace.jsonl. Console progress is concise; events.jsonl retains full run events.

Provider support

The provider adapter supports:

LLM_PROVIDER	Credentials	Live-search mechanism
openai	OPENAI_API_KEY	OpenAI Responses API web_search_preview
openrouter	OPENROUTER_API_KEY	OpenAI-compatible client at OPENROUTER_BASE_URL with OpenRouter server tool openrouter:web_search
anthropic	ANTHROPIC_API_KEY	Anthropic web-search tool

OpenRouter configuration is documented in .env.example:

```
LLM_PROVIDER=openrouter
OPENROUTER_API_KEY=
OPENROUTER_BASE_URL=https://openrouter.ai/api/v1
OPENROUTER_MODEL=openai/gpt-4.1-mini
OPENROUTER_MAX_TOKENS=1000
```

OPENROUTER_MAX_TOKENS is explicit because OpenRouter otherwise reserves the selected model's much larger default maximum, which can exceed a restricted key's limit before a request starts. The default is intentionally small for Stage 0; it can be raised after the account has adequate credit.

Guardrails and auditability

- No data is generated when a provider/search request fails. Failures are logged and the run stops safely.
- Pydantic validates all structured stage outputs. A validation failure triggers one corrected-output retry.
- Markdown-fenced JSON is normalized before validation; this accepts presentation formatting without accepting invalid JSON.
- Contacts require a real two-part name or NOT_FOUND; not_found contacts cannot carry an email.
- Email checks enforce word limits, banned phrases, placeholder detection, allowed FlytBase proof names, evidence presence, and a company/contact swap test.
- Provider failure text is redacted before it is saved to avoid recording bearer credentials or provider key-management identifiers.
- .env is ignored by Git and is never printed by the agent.

Tests

The offline suite does not call the network. It covers:

- source URL schema validation;
- banned phrase and placeholder detection;
- generic-email swap testing;
- NOT_FOUND contact rules;
- OpenRouter client/base URL/token-budget configuration;
- fenced JSON normalization;
- JSON artifact writing;

- parsing the required seven-item ICP rubric from the canonical document.

Latest local result: 9 passed.

Live-run evidence

Run	Result
runs/20261008T030644	Identified OpenRouter's implicit 65,536-token request reservation.
runs/20261008T031146	Live web research completed; surfaced a fenced-JSON and artifact-writer defect.
runs/20261008T031411	Created a live-source ICP artifact, then it was superseded because its rubric did not match docs/ICP.md exactly.
runs/20261008T031950	Final corrected code reached OpenRouter but received definitive HTTP 402 due to the account having no purchased credits.

The corresponding failures.log, events.jsonl, and trace.jsonl files are retained in each run folder. docs/FAILURES_AND_FIXES.md contains only real observed failures.

Remaining issue

The current OpenRouter account/key has no billable API credit balance. The provider's final response was HTTP 402, stating that the account had never purchased credits. This prevents any provider request, web search, or fresh Stage 0 artifact from completing. It is an external account/billing issue rather than a local code issue.

No mock ICP, accounts, contacts, research, emails, or sources were written to work around this block.

Resume instructions

1. Fund the configured OpenRouter account or replace the key with one on a funded account.
2. Confirm .env has the OpenRouter variables above.
3. Run the small live proof:

```
powershell
```

```
python agent/run.py --brief brief.yaml --max-accounts 2 --contacts-per-account 1 --only-stage 0
```

4. Inspect the resulting runs/<timestamp>/icp_profile.json, trace.jsonl, and failures.log.
5. When Stage 0 succeeds, run the complete pipeline:

```
powershell
```

```
python agent/run.py --brief brief.yaml --max-accounts 3 --contacts-per-account 1
```

The full run should be attempted only with enough provider budget for multiple web-search and QA calls. Any future real failure is automatically retained in that run's failures.log.